

Logging using SLF4J

Exercise 1: Logging Error Messages and Warning Levels

Task: Write a Java application that demonstrates logging error messages and warning levels using SLF4J.

Solution:

```
<dependency>
  <groupId>org.slf4j</groupId>
  <artifactId>slf4j-api</artifactId>
  <version>2.0.17</version>
</dependency>
```

```
package com.example;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class LoggingExample {

    private static final Logger logger =
        LoggerFactory.getLogger(
            LoggingExample.class
        );

    public static void main(String[] args) {

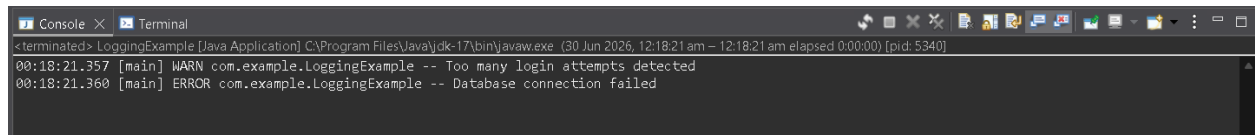
        int loginAttempts = 4;
        boolean serverConnected = false;

        if(loginAttempts > 3)
        {
            logger.warn(
                "Too many login attempts detected"
            );
        }

        if(!serverConnected)
        {
            logger.error(
                "Database connection failed"
            );
        }

    }
}
```

Output:

A screenshot of a Windows console window titled "Console X" and "Terminal". The window shows the output of a Java application. The first line is a status message: "<terminated> LoggingExample [Java Application] C:\Program Files\Java\jdk-17\bin\javaw.exe (30 Jun 2026, 12:18:21 am - 12:18:21 am elapsed 0:00:00) [pid: 5340]". The second line is a warning message: "00:18:21.357 [main] WARN com.example.LoggingExample -- Too many login attempts detected". The third line is an error message: "00:18:21.360 [main] ERROR com.example.LoggingExample -- Database connection failed".

```
<terminated> LoggingExample [Java Application] C:\Program Files\Java\jdk-17\bin\javaw.exe (30 Jun 2026, 12:18:21 am - 12:18:21 am elapsed 0:00:00) [pid: 5340]
00:18:21.357 [main] WARN com.example.LoggingExample -- Too many login attempts detected
00:18:21.360 [main] ERROR com.example.LoggingExample -- Database connection failed
```

Exercise 2: Parameterized Logging

Task: Write a Java application that demonstrates parameterized logging using SLF4J.

Solution:

ParameterizedLogging.java

```
package com.example;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class ParameterizedLogging {

    private static final Logger logger =
        LoggerFactory.getLogger(
            ParameterizedLogging.class
        );

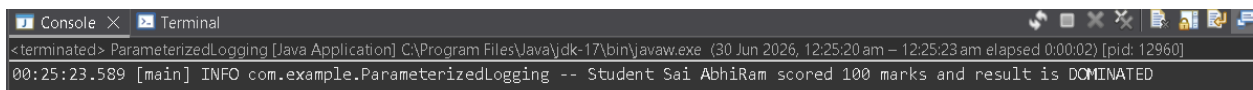
    public static void main(String[] args) {

        String student = "Sai AbhiRam";
        int marks = 100;
        String result = "DOMINATED";

        logger.info(
            "Student {} scored {} marks and result is {}",
            student,
            marks,
            result
        );

    }
}
```

Output:



```
Console X Terminal
<terminated> ParameterizedLogging [Java Application] C:\Program Files\Java\jdk-17\bin\javaw.exe (30 Jun 2026, 12:25:20 am – 12:25:23 am elapsed 0:00:02) [pid: 12960]
00:25:23.589 [main] INFO com.example.ParameterizedLogging -- Student Sai AbhiRam scored 100 marks and result is DOMINATED
```